



**POLITECNICO
MILANO 1863**

**SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE**

EXECUTIVE SUMMARY OF THE GROUP PROJECT

Kira - ISR Unmanned Aerial System

AERONAUTICAL ENGINEERING - CORSI DI PROGETTO DI VELIVOLI

Author: FILIPPO COACCI

Advisors: PROF. LORENZO TRAINELLI, PROF. CARLO EMANUELE DIONIGI RIBOLDI

Academic year: 2024-2025

1. System at a Glance

Mission & Value Proposition

KIRA is a fixed-wing unmanned aerial system designed for persistent ISR missions with counter-UAS (C-UAS) capability against drone threats of all dimensions (down to MAV-class). The platform is sized to provide long on-station time, modular payload integration, and expeditionary operations from semi-prepared sites, supported by a mobile GCS concept.



Figure 1: KIRA ISR/C-UAS Platform.

Core Capabilities

1. Persistent surveillance of stationary or moving targets (ISR), day/night and in degraded visibility (sensor-kit dependent).

2. Detection, classification, and tracking of small aerial threats. Soft-kill (EW) and cueing to GCS for coordinated response (payload/ROE dependent).
3. Rapid deployment with minimal infrastructure such as short-field grass operations.
4. Civil-logistics deployable: the full system is transportable using two standard commercial vans (e.g., Mercedes Sprinter class), enabling rapid fielding without dedicated military transport.

System Architecture

- **Air vehicle:** high aspect ratio fixed-wing for endurance, pusher configuration for payload FOV, tricycle gear for sensor ground clearance and rough-field handling.
- **Payload bay:** modular mission kit (radar + EO/IR + EW, optional kits).
- **GCS:** mobile GCS concept, adaptable to single-UAV or dual-UAV coordinated orbits.
- **Deployment concept:** two-van package — Van #1 carries two modular/detached air vehicles; Van #2 carries the mobile GCS (operators, comms, spares), enabling rapid setup and relocation.
- **Dual-use certification strategy:** civil baseline via EASA opt-in approach, with military alignment/validation (ARMAERO / STANAG 4671) planned.

Key Specifications

Mass properties

MTOM	640 kg
OEM	370 kg
Fuel Mass	145 kg
PL	125 kg

Geometry

Wing span b	10.0 m
Wing area S	6.9 m ²
Aspect ratio AR	15 -

Performance (indicative)

Stall speed V_{st}	25 m/s
Loiter speed V_{loi}	40 m/s
Range speed V_{range}	45 m/s
Max speed V_{max}	115 m/s
Max ROC	10 m/s
Ceiling	~10,500 m
Takeoff (grass)	< 400 m
Landing (grass)	< 400 m

Electrical Power Budget

APU available power	11 kW
---------------------	-------

Baseline electrical loads

Radar	0.3 kW
EO/IR	0.2 kW
Jammer / EW	0.5 kW
Avionics	1.5 kW
Actuators (peak)	0.5 kW

Payload systems (max combined)	~ 3.0 kW
--------------------------------	----------

PL vs Endurance

The maximum loiter endurance of 25 h is achieved with full payload (125 kg) in the baseline ISR configuration. The total is composed of 24 h mission + 1 h contingency/reserve, for eventual emergency scenarios.

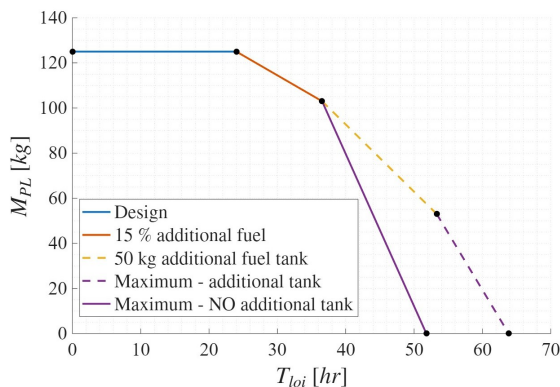


Figure 2: PL vs Endurance Diagram.

Note: The maximum *Range* capability of Kira is ~ 2'200 nm with full payload and the *Ferry-Range* is > 4'400 nm with no on board payload.

CONOPS

The concepts of operations comprehend the use of either one UAV at lower orbit altitude or 2 UAVs at high orbit altitude.

- **Single-UAV:** Loiter altitude of 1'300 m AGL and orbi radius of 1'950 m.
- **Dual-UAV Coordinated:** Loiter altitude of 2'500 m AGL and orbit radius of 2'400 m. The two UAVs ensure persistent surveillance and threat coverage on opposing 180° orbits around the target.

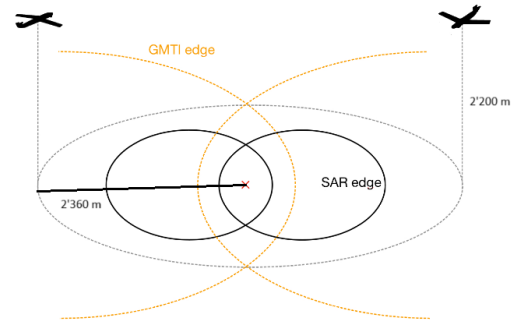


Figure 3: Schematic Dual - UAV CONOPS.

Threat Engagement Timeline

At representative threat kinematics, this indicative timeline supports engagement at stand-off distance from the protected area (buffered), subject to payload selection and rules of engagement.

Phase	Time [s]
Detection	10.0
Communication	5.0
Neutralisation	4.0
Total	19

Assumptions / Scope

All figures are sizing-level / preliminary-design estimates; performance depends on final aero/prop integration and mission kit definition.

2. Design Summary

Design Drivers

- **Persistence at full mission payload** (125 kg) drives an endurance-optimised aerodynamic configuration (high- AR wing, loiter-centric sizing) and mission-level mass/power margins.
- **Expeditionary footprint** drives low wing loading and rough-field performance, enabling operations from grass/semi-prepared strips with limited infrastructure.
- **Payload modularity and growth** are enabled by a capped payload electrical budget (≤ 3 kW combined) while retaining headroom from the on-board APU (11 kW available) for transients, thermal management, and future kits.

Aerodynamic Sizing

The selected W/S and high-lift configuration ensure low-speed handling and short-field margins, while the loiter L/D supports the required endurance at full payload. The endurance-driven high- AR wing is structurally sized at preliminary level to limit the associated weight penalty. The optimization procedure and design iterations led to a optimal wing loading of about 90 kg/m^2 and a lift-to-drag ratio during loiter greater than 20.

Propulsion & Actuators

A pusher installation is selected primarily for payload FOV preservation and improved stability. Variable pitch enables a take-off thrust margin while maintaining loiter efficiency.

Powerplant	Rotax 914
Max shaft power	$\sim 85 \text{ kW}$
Typical loiter power	$\sim 16 \text{ kW}$
Propeller	3-blade, variable pitch
Propeller diameter	$\sim 1.7 \text{ m}$

Primary flight control actuation (servomotors) is fully electric, simplifying integration and maintainability. The combined control surface actuation peak electrical demand is $\sim 0.5 \text{ kW}$ (short duty-cycle), already included in the worst-case electrical assessment.

Stability & CG Management

An optimised and redundant cross-feed fuel consumption strategy is adopted to keep the CG inside the safe envelope throughout the 24 h mission. CG excursion remains limited and bounded between the forward position, happening at $t = 3 \text{ h}$ into mission, corresponding to a static margin (SM) of about 12% and the aft position at the end of the mission (SM $\sim 7\%$).

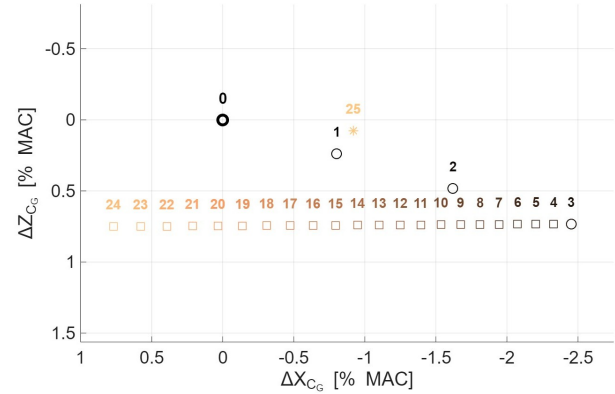


Figure 4: Mission CG Excursion (hourly).

Structural Sizing

Structural sizing is developed against *STANAG 4671 (USAR)* as provisional design basis: the $V-n$ and gust envelopes define the critical manoeuvre and gust load cases (1.5 safety factor). A multidisciplinary loop combines composite sizing with stick-based aero/structural modelling and VLM load extraction. Results are combined into shear-bending and bending-torsion envelopes at the wing root and wing-fuselage interface. Composite lay-ups are iterated to satisfy strength margins (Tsai-Hill) while minimising mass.

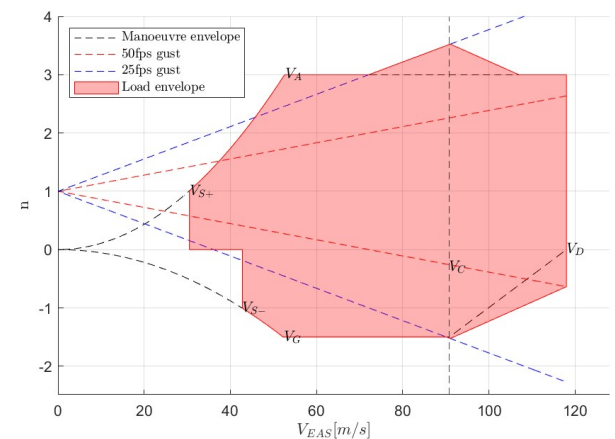


Figure 5: $V-n$ Gust and Load Envelope.

3. Cost and Roadmap

Market Fit & Positioning

KIRA targets customers requiring **mobile** ISR + C-UAS coverage where fixed ground systems are constrained by terrain, infrastructure, or operational mobility. It occupies the **mid-budget** niche by combining: **persistence at full payload** (25 h with 125 kg), **modular radar+EO/IR+EW integration**, **civil-logistics deployability** (expeditionary footprint).

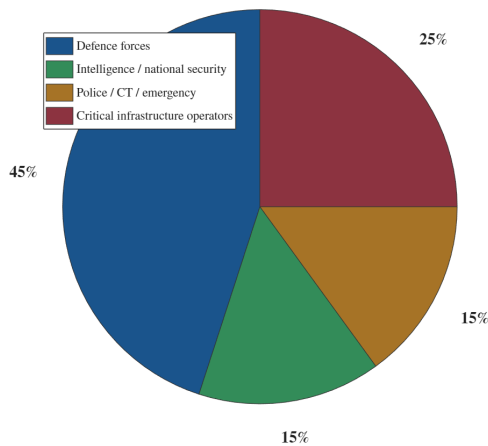


Figure 6: Target Customer Segments.

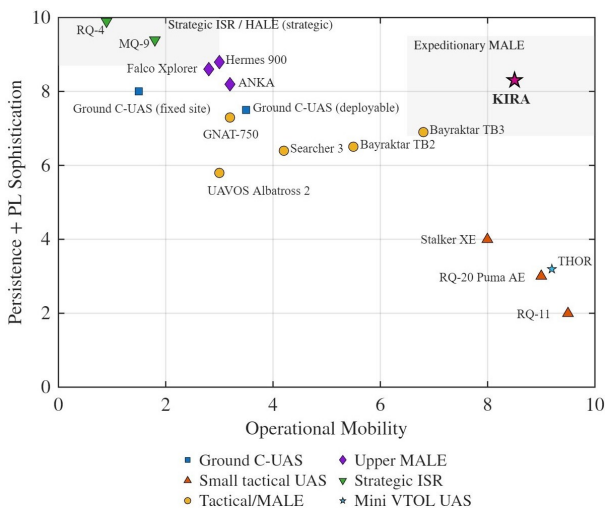


Figure 7: KIRA Niche Market Position.

Acquisition Configurations

- **Main package:** 2×UAV + mobile GCS (continuous coverage via coordinated operations).
- **Secondary package:** 1×UAV + mobile GCS (single-platform ISR/C-UAS support).
- **Optional:** mission-based **service/leasing**

model for customers avoiding full procurement.

Cost & Commercial Model

Preliminary bottom-up estimate.

Item	Value USD
Single UAV total cost	3.81 M
Mobile GCS cost	1.45 M
Main UAS cost	9.08 M
Secondary UAS cost	5.27 M
24 h mission leasing (indicative)	0.14 M

Indicative price

Main package price	18.16 M
Secondary package price	10.54 M
Per-mission leasing price (24 h)	0.24 M

Dual-Use Compliance Strategy

A civil-first certification strategy via EASA (opt-in) was selected to ensure international recognition and exportability, with subsequent military validations as required. This choice drives key architectural decisions, including conservative stability margins, airworthiness-driven structural sizing (STANAG 4671 benchmark), defined operational envelopes, and staged verification. The system is designed for grass-runway operations of 400 m, high crosswind tolerance, and incorporates redundancy on critical elements (flight-control actuation, avionics, and radar payloads).

Roadmap & Key Risks

Maturity: TRL 2–3 (concept and analytical proof-of-concept; no hardware testing performed). Progression to higher TRLs is driven by interface freeze and payload mock-ups (TRL 4), avionics and flight-control HIL with EMI/EMC ground testing (TRL 5), and full-scale prototype manufacturing with incremental flight-envelope expansion (TRL 6). The primary program risk is composite airframe manufacturing cost and schedule, mitigated through early structural detailing, laminate optimisation, and tooling strategy; additional risks include payload integration, EMI/EMC, and data-link robustness, addressed through early interface definition, bench testing, and degraded-mode autonomy.